



EXAM PAPERS PRACTICE

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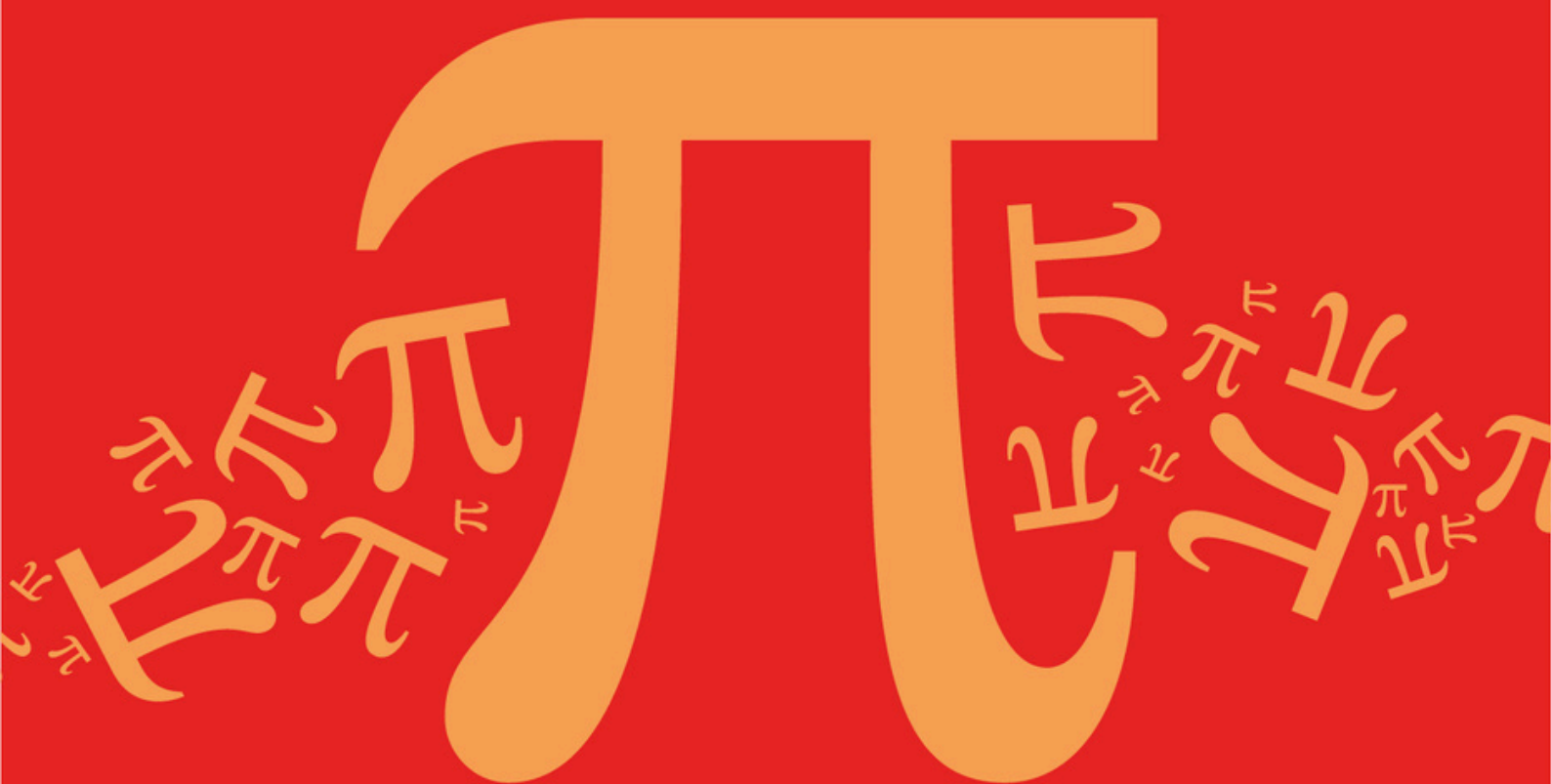
Practice questions created by actual
examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and
thoroughly prepare you

AQA A Level Further Mathematics (7367)



Topic

C: Matrices

Sub topic

C10: Eigenvalues and Eigenvectors

Mark Scheme

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Topic

C: Matrices

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C10: Eigenvalues and Eigenvectors

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Forms appropriate equation using $\mathbf{M}\lambda = \lambda\mathbf{v}$	AO1.1a	M1	$\begin{bmatrix} -1 & 2 & -1 \\ 2 & 2 & -2 \\ -1 & -2 & -1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 4 \begin{bmatrix} a \\ b \\ c \end{bmatrix}$
	Eliminates one variable	AO1.1a	M1	
	Deduces a correct eigenvector	AO2.2a	A1	
				$\begin{aligned} -5a + 2b - 2c &= 0 \\ 2a - 2b - 2c &= 0 \\ -a - 2b - 5c &= 0 \\ 3a + 3c &= 0 \end{aligned}$ $\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$ eigenvector is
(b)	Forms the characteristic equation of $\mathbf{M}$	AO3.1a	M1	$\begin{vmatrix} -1-\lambda & 2 & -1 \\ 2 & 2-\lambda & -2 \\ -1 & -2 & -1-\lambda \end{vmatrix} = 0$ $(-1-\lambda)[(2-\lambda)(-1-\lambda) - 4]$ $-2(-2-2\lambda-2) - 1(-4+2-\lambda) = 0$ $-\lambda^3 + 12\lambda + 16 = 0$ $(4-\lambda)(\lambda^2 + 4\lambda + 4) = 0$ $-(\lambda+2)(\lambda-4)(\lambda+2) = 0$ Eigenvalues are 4, -2, -2 $\begin{bmatrix} -1 & 2 & -1 \\ 2 & 2 & -2 \\ -1 & -2 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = -2 \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ $x + 2y - z = 0$
	Obtains the correct characteristic equation – unsimplified	AO1.1b	A1	
	Obtains roots and identifies them as eigenvalues for ‘their’ characteristic equation	AO1.1b	A1F	
	Forms an appropriate matrix equation using the eigenvalue -2 FT ‘their’ eigenvalue	AO3.1a	M1	
	Expands and simplifies to obtain a single equation in x, y and z FT ‘their’ matrix equation provided both M1 marks have been awarded	AO1.1b	A1F	
	Correctly deduces two linearly independent eigenvectors CAO	A02.2a	A1	
	Correctly identifies that the matrix $\mathbf{D}$ must include 4	AO1.2	B1F	

and 'their' other eigenvalue(s)			$\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$
Correctly identifies the corresponding U matrix from 'their' eigenvectors	AO1.1b	A1F	$\mathbf{D} = \begin{bmatrix} 4 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$ $\mathbf{U} = \begin{bmatrix} 1 & 1 & -1 \\ 2 & 0 & 1 \\ -1 & 1 & 1 \end{bmatrix}$
Total 11 marks			

Q2.

(a)
$$\begin{bmatrix} -a & 0 & a \\ 0 & 6 & 0 \\ a & 0 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 6 \\ 0 \end{bmatrix}$$

M1

$\Rightarrow \lambda_1 = 6$

A1

(b)
$$\begin{bmatrix} -a & 0 & a \\ 0 & 6 & 0 \\ a & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} a \\ 0 \\ a+4 \end{bmatrix}$$

M1

$$\begin{bmatrix} a \\ 0 \\ a+4 \end{bmatrix} = \begin{bmatrix} \lambda_2 \\ 0 \\ 2\lambda_2 \end{bmatrix}$$

i component $\Rightarrow a = \lambda_2 \quad \otimes$

k component $\Rightarrow a + 4 = 2\lambda_2$

using $\otimes$, $a + 4 = 2a$

Eliminating λ_2

m1

$4 = a$

Value of a obtained

A1

(c) Let $\mathbf{v}_3 = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

$$\begin{bmatrix} -4 & 0 & 4 \\ 0 & 6 & 0 \\ 4 & 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = -6 \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Substitute 'their value of a' and attempt to get a system of equations.

M1

$$\Rightarrow -4x + 4z = -6x \Rightarrow x + 2z = 0$$

$$6y = -6y \Rightarrow y = 0$$

$$[4x + 2z = -6z \Rightarrow x + 2z = 0]$$

Both equations "correct" FT their a

A1F

$$\mathbf{v}_3 = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} \text{ (or equivalent)}$$

A1cao

3

Alternative:

$$\begin{bmatrix} -4 & 0 & 4 \\ 0 & 6 & 0 \\ 4 & 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = -6 \begin{bmatrix} x \\ y \\ z \end{bmatrix} \Rightarrow \begin{bmatrix} 2 & 0 & 4 \\ 0 & 12 & 0 \\ 4 & 0 & 8 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$r_3 \rightarrow r_3 - 2r_1 \quad \begin{bmatrix} 2 & 0 & 4 \\ 0 & 12 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Row operations

(M1)

$$\Rightarrow y = 0 \quad x = -2z$$

"correct" FT their a

(A1F)

$$\mathbf{v}_3 = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} \text{ (or equivalent)}$$

(A1cao)

(3)

(d) $D = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & -6 \end{bmatrix}$

Diagonal matrix using -6 and "their 4" and "their 6"

B1F

$U = \begin{bmatrix} 0 & 1 & -2 \\ 1 & 0 & 0 \\ 0 & 2 & 1 \end{bmatrix}$

FT their non-zero v_3 in U

U correct and corresponding to D

A1cao

3

[11]

Q3.

(a) $\begin{vmatrix} 1-\lambda & 1 & 2 \\ 0 & 2-\lambda & 2 \\ -1 & 1 & 3-\lambda \end{vmatrix} = 0$

$(1-\lambda) \begin{vmatrix} 2-\lambda & 2 \\ 1 & 3-\lambda \end{vmatrix} - \begin{vmatrix} 1 & 2 \\ 2-\lambda & 2 \end{vmatrix} = 0$

Correct row / column expansion of $|M - \lambda I| = 0$

M1

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$(1-\lambda)(\lambda-4)(\lambda-1) - 2(\lambda-1) = 0$

$(1-\lambda)[(2-\lambda)(3-\lambda) - 2] - [2 - 2(2-\lambda)] = 0$

Correct expansion of 2 by 2 determinants

- dependent on first M1

m1

$[\lambda - 1][-\lambda^2 + 5\lambda - 6] = 0$

or $-\lambda^3 + 6\lambda^2 - 11\lambda + 6 = 0$

A1

$-(\lambda-1)(\lambda-2)(\lambda-3) = 0$

Attempt to show $\lambda = 2$ is an eigenvalue

M1

$\lambda = 1, 2 \text{ or } 3$

Fully correct eigenvalues - CAO

A1

5

(b)
$$\begin{bmatrix} -1 & 1 & 2 \\ 0 & 0 & 2 \\ -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Attempt to solve $(M - 2I) \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$

M1

$$\begin{array}{l} \Rightarrow 2z = 0 \\ -x + y + z = 0 \end{array} \quad \left. \vphantom{\begin{array}{l} \Rightarrow 2z = 0 \\ -x + y + z = 0 \end{array}} \right\} \begin{array}{l} \Rightarrow z = 0 \\ x = y \end{array}$$

Both relationships obtained (can be unsimplified)

A1

$\Rightarrow \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ is an eigenvector

Stated clearly; accept multiples

A1

3

(c) (i)
$$\begin{pmatrix} 2 & 2 & -3 \\ 2 & 2 & 3 \\ -3 & 3 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \\ 0 \end{pmatrix} = 4 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

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Attempt at $N \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$

M1

$\Rightarrow \lambda = 4$ is an eigenvalue

A1

2

(ii) Eigenvector = $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$

Accept multiples – part (b) must be correct

B1

Eigenvalue = $2 \times 4 = 8$

M1 for product of relevant eigenvalues

M1A1

3

Alternative:

$$\text{Eigenvector} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

Accept multiples – part b) must be correct

(B1)

$$\begin{pmatrix} 1 & 1 & 2 \\ 0 & 2 & 2 \\ -1 & 1 & 3 \end{pmatrix} \begin{pmatrix} 2 & 2 & -3 \\ 2 & 2 & 3 \\ -3 & 3 & 3 \end{pmatrix} = \begin{pmatrix} -2 & 10 & 6 \\ -2 & 10 & 12 \\ -9 & 9 & 15 \end{pmatrix}$$

$$\begin{pmatrix} -2 & 10 & 6 \\ -2 & 10 & 12 \\ -9 & 9 & 15 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 8 \\ 8 \\ 0 \end{pmatrix} = 8 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

Multiplies to get MN and attempts to find eigenvalue

(M1)

So eigenvalue is 8

Correct eigenvalue

(A1)

(3)

[13]

Q4.

(a) Char. Eqn. is $\lambda^2 - 8\lambda - 9 = 0$

Attempted

M1

Quadratic solved to get two roots

dM1

$$\Rightarrow \lambda = 9, -1$$

A1

Subst^g. back λ (at least once):

M1

$$\lambda = 9 \Rightarrow -x + y = 0 \Rightarrow \lambda = 9 \text{ has evecs. } \alpha \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

any $\alpha \neq 0$

$$\lambda = -1 \Rightarrow x + y = 0 \Rightarrow \lambda = -1 \text{ has evecs. } \beta \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

any $\beta \neq 0$

A1

6

(b) (0, 0)

B1

1

[7]

Q5.

(a) (i) $\mathbf{A}^T = \begin{bmatrix} k & 2 & 1 \\ 1 & k & 2 \\ 2 & 1 & k \end{bmatrix}$

B1

Good multiplication attempt at $\mathbf{A} \mathbf{A}^T$

If they multiply $\mathbf{A}^T \mathbf{A}$ instead, they can score B1 M1 A0 A0 A1 A1

A1A1

M1

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$$= \begin{bmatrix} k^2+5 & 3k+2 & 3k+2 \\ 3k+2 & k^2+5 & 3k+2 \\ 3k+2 & 3k+2 & k^2+5 \end{bmatrix}$$

Main diagonal correct

A1

All others correct

A1

$$k = -\frac{2}{3} \quad m = 5\frac{4}{9} \text{ or } \frac{49}{9}$$

6

(ii) $\begin{bmatrix} -\frac{2}{3} & 1 & 2 \\ 2 & -\frac{2}{3} & 1 \\ 1 & 2 & -\frac{2}{3} \end{bmatrix}^{-1} = \frac{9}{49} \begin{bmatrix} -\frac{2}{3} & 2 & 1 \\ 1 & -\frac{2}{3} & 2 \\ 2 & 1 & -\frac{2}{3} \end{bmatrix}$

ft their m and their k

B1

Accept $\frac{9}{49} \begin{bmatrix} k & 2 & 1 \\ 1 & k & 2 \\ 2 & 1 & k \end{bmatrix}$ since the value of k is now known,
but not just $\frac{9}{49} \mathbf{A}^T$

Decimal version (correct to at least 3sf) is also ok:

$$\begin{bmatrix} -0.122 & 0.367 & 0.184 \\ 0.184 & -0.122 & 0.367 \\ 0.367 & 0.184 & -0.122 \end{bmatrix}$$

1

(b) (i) $\det \mathbf{A} = k^3 - 6k + 9$

Good attempt (cubic)

M1

A1

2

(ii) $\det \mathbf{A} = (k + 3)(k^2 - 3k + 3)$

Factorisation attempt

M1

$k = -3$

A1

$\Delta = 9 - 12$

B1

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Special Cases

$k = -3$ but no working

(B1)

For all 3 roots, -3 and $\frac{3+i\sqrt{3}}{2}$ given
with no supporting working

(B3)

3

(iii) Replacing k by $(k - 7)$

NOT just in the determinant form (ie starting again)

M1

$k = 4$ (however obtained)

ft their previous $k + 7$

A1

2

Q6.

(a) Attempt at Char.Eqn.

M1

$$\lambda^3 - 9\lambda^2 + 24\lambda - 16 = 0$$

One each following coefft.

A3,2,1

Attempt at (at least) one linear factor

M1

$$(\lambda - 1)(\lambda - 4)^2 = 0$$

A1

$$\lambda = 1, 4, 4$$

A1

7

(b)

$$\begin{aligned} 2x - y + z &= 0 \\ \lambda = 1 \Rightarrow -x + 2y + z &= 0 \Rightarrow \alpha \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \\ x + y + 2z &= 0 \end{aligned}$$

Substⁿ. back and solving

M1

A1

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$$\lambda = 4 \Rightarrow x + y - z = 0$$

B1

Choosing any two independent vectors satisfying this

M1

$$\text{E.g.s: } \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

A1

5

$$(c) \quad \alpha \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \equiv \text{a line of invariant points}$$

Invariant line

M1

LoIPs

A1

e.g. $\alpha \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \equiv \text{an invariant plane}$

B1

3

[15]

Q7.

- (a) (i) Characteristic eqn $\lambda^2 - 9\lambda + 14 = 0$
M1A0 if num error(s) made

M1A1

$\lambda = 2, 7$

A1

Substituting back for at least one eval

for $\lambda = 2, -x + 3y = 0$ or

for $\lambda = 7, -2x + y = 0$

m1

evecs $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$
or non-zero multiples

A1A1

6

(ii) $\mathbf{U} = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}, \mathbf{D} = \begin{bmatrix} 2 & 0 \\ 0 & 7 \end{bmatrix}$

Columns of $\mathbf{U}$ and $\mathbf{D}$ are interchangeable,
 but must match;

ft wrong answers in (i)

B1FB1F

$\mathbf{U}^{-1} = \frac{1}{5} \begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix}$

1/det $\mathbf{U}$

B1F

adjoint matrix;

ft incorrect $\mathbf{U}$ (provided $\det \neq 0$)

B1F 4

- (b) (i) evals of $\mathbf{M}^3$ are λ^3, μ^3

B1

since $\mathbf{M}^3 = \mathbf{U} \mathbf{D}^3 \mathbf{U}^{-1}$

E1 2

- (ii) evects of $\mathbf{M}^3$ are $\mathbf{v}_1$ and $\mathbf{v}_2$

B1 1

[13]

Q8.

- (a) (i) $R_1' = R_1 + R_2$ or $R_2' = R_2 + R_1$

M1

leading to $R_{1/2} = (4 - q \quad 4 - q \quad 0)$

A1 2

(ii)
$$\Delta = (4 - q) \begin{vmatrix} 1 & 1 & 0 \\ -12 & -1 - q & -7 \\ 6 & 6 & 10 - q \end{vmatrix}$$

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$$= (4 - q) \begin{vmatrix} 0 & 1 & 0 \\ q - 11 & -1 - q & -7 \\ 0 & 6 & 10 - q \end{vmatrix}$$

By $C_1' = C_1 - C_2$ (eg)

M1

$$= (4 - q)(q - 11) \begin{vmatrix} 0 & 1 & 0 \\ 1 & -1 - q & -7 \\ 0 & 6 & 10 - q \end{vmatrix}$$

2nd linear factor correct

A1

$$= (4 - q)(q - 11) \times \dots$$

Full attempt at 3rd factor

M1

$$= (4 - q)(q - 11)(q - 10)$$

A1

4

(b) (i)
$$\begin{bmatrix} 16 & 5 & 7 \\ -12 & -1 & -7 \\ 6 & 6 & 10 \end{bmatrix} \begin{bmatrix} 2 \\ 5 \\ -7 \end{bmatrix} = \begin{bmatrix} 8 \\ 20 \\ -28 \end{bmatrix}$$

M1A1

$$= 4 \begin{bmatrix} 2 \\ 5 \\ -7 \end{bmatrix} \text{ so that } \lambda = 4$$

A1

3

(ii) $\lambda = 10, 11$ noted or used
ft

$$\begin{aligned} 0 &= 6x + 5y + 7z \\ \lambda = 10 \Rightarrow 0 &= -12x - 11y - 7z \\ 0 &= 6x + 6y \end{aligned}$$

B1

B1

Using $y = -x$ to get $z \sim x/y$

M1

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Eigenvector $(-7, 7, 1)$

Any non-zero multiple will do

A1

$$\begin{aligned} 0 &= 5x + 5y + 7z \\ \lambda = 11 \Rightarrow 0 &= -12x - 12y - 7z \\ 0 &= 6x + 6y - z \end{aligned}$$

Either $y = -x$ or $z = 0$

M1

Eigenvector $(1, -1, 0)$

Any non-zero multiple will do

A1

7

(c)
$$\frac{x}{2} = \frac{y}{5} = \frac{z}{-7}, \quad \frac{x}{-7} = \frac{y}{7} = z$$

Any line eqns.

M1

or $x = -y, z = 0$

any one ft correct

A1

2

[18]

Q9.

(a) (i) Evals $\lambda = 27$ 1
Both

B1

Evecs $(\alpha) \begin{bmatrix} 4 \\ 1 \end{bmatrix}$ and $(\beta) \begin{bmatrix} -1 \\ 3 \end{bmatrix}$

Correctly matched up with evals
(look out for $\lambda_1, \mathbf{v}_1$ notations)

B1B1B1

4

(ii) $y = -3x$
Ft $4y = x$ if evecs mis-matched

B1ft

from $\lambda = 1$

Must say why they have chosen this one

B1

2

(b) $\mathbf{U}^{-1} = \frac{1}{13} \begin{bmatrix} 3 & 1 \\ -1 & 4 \end{bmatrix}$
Det; mtx

B1B1

$\mathbf{M} = \mathbf{U} \mathbf{D} \mathbf{U}^{-1}$

Including attempt to multiply
(at least $\mathbf{U} \mathbf{D} \dots$)

M1

$$\begin{aligned}
 &= \frac{1}{13} \begin{bmatrix} 4 & -1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 27 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ -1 & 4 \end{bmatrix} \\
 &= \frac{1}{13} \begin{bmatrix} 4 & -1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 81 & 27 \\ -1 & 4 \end{bmatrix}
 \end{aligned}$$

*Ft incorrect/missing $\mathbf{U}^{-1}$ for one product;
ignore missing $\frac{1}{13}$ until the end*

A1

$$\begin{aligned}
 &\text{or } \frac{1}{13} \begin{bmatrix} 108 & -1 \\ 27 & 3 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ -1 & 4 \end{bmatrix} \\
 &= \begin{bmatrix} 25 & 8 \\ 6 & 3 \end{bmatrix}
 \end{aligned}$$

CAO

A1

5

(c) $\mathbf{M}^n = \mathbf{U} \mathbf{D}^n \mathbf{U}^{-1}$

Including attempt to multiply

M1

$$\mathbf{D}^n = \begin{bmatrix} 27^n & 0 \\ 0 & 1 \end{bmatrix}$$

B1

$$\mathbf{M}^n(1, 1) = \frac{1}{13} (12 \times 27^n + 1)$$

A1

So $4 \times 3 \times 3^{3n} + 1 = 4 \times 3^{3n+1} + 1$ div. by 13

*Legitimately so from their working, from
fact that the element is an integer*

E1

Since $\mathbf{M}$ has all integer elements, each
element of $\mathbf{M}^n$ is an integer also

Explaining why it must be an integer

E1

5

[16]

Q10.

(a) Subst^g. $\lambda = -1$ into $\det(\mathbf{M} - \lambda \mathbf{I}) = 0$

Or $\mathbf{M} \mathbf{x} = -\mathbf{x}$ etc.

M1

Solving between $x + y + z = 0$
and $x + y + 2z = 0$

dM1

Eigenvector(s) $\alpha \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$
Any non-zero α will suffice

A1

3

(b) Attempt at Char. Eqn.

M1

$\lambda^3 - 5\lambda^2 - 5\lambda + 1 = 0$
Each coeff. (not the λ^3)

A1 $\times$ 3

Use of division/factor theorem etc.
With/without $(\lambda + 1)$ factor

M1

$(\lambda + 1)(\lambda^2 - 6\lambda + 1)$

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A1

Solving remaining quadratic factor

M1

$\lambda_{2,3} = 3 \pm 2\sqrt{2}$
CAO simplest exact form

A1

8

[11]

Q11.

(a) $\text{Det}(\mathbf{M}) = 1 \Rightarrow$ Area invariant under T
2nd B1 ft ref. "area"

B1B1

2

(b) Char. Eqn. $\lambda^2 - 2\lambda + 1 = 0$

M1

$$\Rightarrow \lambda = 1 \text{ (twice)}$$

A1

Subst^g. their λ back to find an evect:

M1

$$\alpha \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Any (non-zero) α

A1

(Since $\lambda = 1$) this represents a line of inv. pts.

ft if $\lambda \neq 1$

B1

5

$$(c) \quad \begin{bmatrix} -1 & 4 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} x \\ \frac{1}{2}x + k \end{bmatrix} = \begin{bmatrix} x + 4k \\ \frac{1}{2}x + 3k \end{bmatrix}$$

M1A1

Verifying that $y' = \frac{1}{2}x' + k$

Be convinced AG

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A1

3

[10]

Q12.

$$(a) \quad (i) \quad \mathbf{D} = \begin{bmatrix} 3 & 0 \\ 0 & -3 \end{bmatrix} \quad \mathbf{U} = \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix}$$

$\mathbf{D}, \mathbf{U}$ (alt. choices ok)

B1B1

$$\mathbf{U}^{-1} = \frac{1}{2} \begin{bmatrix} 4 & -1 \\ -2 & 1 \end{bmatrix}$$

ft 1st B1 provided $\det \neq 0$

B1

ft 2nd B1 in non-trivial cases

B1

4

(ii)
$$\mathbf{M} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & -3 \end{bmatrix} \begin{bmatrix} 4 & -1 \\ -2 & 1 \end{bmatrix}$$

Some attempt at mtx. multn.

M1

$$= \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 12 & -3 \\ 6 & -3 \end{bmatrix} \text{ or}$$

First multn. correct ft

A1

$$\frac{1}{2} \begin{bmatrix} 3 & -3 \\ 6 & -12 \end{bmatrix} \begin{bmatrix} 4 & -1 \\ -2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 9 & -3 \\ 24 & -9 \end{bmatrix}$$

Ft missing $\frac{1}{2}$ only

A1

3

(b) (i) When n even, $\mathbf{D}^n = \begin{bmatrix} 3^n & 0 \\ 0 & 3^n \end{bmatrix}$

Incl. use in mtx. multn. of form $\mathbf{U} \mathbf{D}^n \mathbf{U}^{-1}$

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M1

$$\mathbf{M}^n = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 4 \cdot 3^n & -3^n \\ -2 \cdot 3^n & 3^n \end{bmatrix} \text{ or}$$

$$\frac{1}{2} \begin{bmatrix} 3^n & 3^n \\ 2 \cdot 3^n & 4 \cdot 3^n \end{bmatrix} \begin{bmatrix} 4 & -1 \\ -2 & 1 \end{bmatrix} \text{ correct}$$

Correct ft

A1

Showing $\mathbf{M}^n = 3^n \mathbf{I}$ legitimately

A1

3

(ii) When n odd, $\mathbf{D}^n = \begin{bmatrix} 3^n & 0 \\ 0 & -3^n \end{bmatrix}$

Incl. use in mtx. multn. of form $\mathbf{U} \mathbf{D}^n \mathbf{U}^{-1}$

M1

$$\mathbf{M}^n = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 4 \cdot 3^n & -3^n \\ 2 \cdot 3^n & -3^n \end{bmatrix} \text{ or } \frac{1}{2} \begin{bmatrix} 3^n & -3^n \\ 2 \cdot 3^n & -4 \cdot 3^n \end{bmatrix} \begin{bmatrix} 4 & -1 \\ -2 & 1 \end{bmatrix} \text{ correct}$$

Correct ft

A1

Showing $\mathbf{M}^n = 3^{n-1} \mathbf{M}$ legitimately

A1

3

[13]

Q13.

(a) $\mathbf{D} = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}, \mathbf{U} = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix},$

B1B1

$$\mathbf{U}^{-1} = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}$$

ft $\mathbf{U}^{-1}$

B1

3

(b) $\mathbf{T}^n = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 2^n & 0 \\ 0 & 2^n \end{bmatrix} \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}$

For $\mathbf{D}^n$ with n even

B1

For use of $\mathbf{U}^{-1} \mathbf{D}^n \mathbf{U}$ form

M1

$$= \begin{bmatrix} 2^n & 2 \times 2^n \\ 2^n & 3 \times 2^n \end{bmatrix} \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}$$

m1A1

$$\text{or } \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 3 \times 2^n & -2 \times 2^n \\ -2^n & 2^n \end{bmatrix} \\ = 2^n \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Shown legitimately

A1

5

Alternative:

$$\mathbf{D}^n = \begin{bmatrix} 2^n & 0 \\ 0 & 2^n \end{bmatrix}$$

For $\mathbf{D}^n$ with n even

(B1)

$$\mathbf{T}^n = \mathbf{U} (2^n \mathbf{I}) \mathbf{U}^{-1}$$

(M1)

$$= 2^n (\mathbf{U} \mathbf{I} \mathbf{U}^{-1})$$

(m2)

$$= 2^n \mathbf{I}$$

Allow $\equiv$ forms such as $3 \cdot 2^n - 2^{n+1}$

(A1)

(5)

[8]

Q14.

(a) (i) $\mathbf{M}^2 = \begin{bmatrix} 2 & -1 & 1 \\ 1 & 0 & 1 \\ 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 & 1 \\ 1 & 0 & 1 \\ 1 & -1 & 2 \end{bmatrix}$

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$$= \begin{bmatrix} 4 & -3 & 3 \\ 3 & -2 & 3 \\ 3 & -3 & 4 \end{bmatrix}$$

A1

$$\begin{aligned} \mathbf{M}^2 + 2\mathbf{I} &= \begin{bmatrix} 4 & -3 & 3 \\ 3 & -2 & 3 \\ 3 & -3 & 4 \end{bmatrix} + \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 6 & -3 & 3 \\ 3 & 0 & 3 \\ 3 & -3 & 6 \end{bmatrix} = 3\mathbf{M} \end{aligned}$$

ie $k = 3$

A1

3

(ii) Multiplying by $\mathbf{M}^{-1}$

M1

to get $\mathbf{M} + 2\mathbf{M}^{-1} = 3\mathbf{I}$
ft

A1

so that $\mathbf{M}^{-1} = \frac{3}{2}\mathbf{I} - \frac{1}{2}\mathbf{M}$
ie $a = -\frac{1}{2}$ and $b = \frac{3}{2}$

A1

3

(b) (i) Char. eqn. is $\lambda^3 - 4\lambda^2 + 5\lambda - 2 = 0$

One A mark for each of the other coefficients

M1A1A1A1

ie $(\lambda - 2)(\lambda - 1)^2 = 0$

Good factorisation attempt

M1

giving $\lambda_1 = 1$ (twice) and $\lambda_2 = 2$

A1

6

(ii) $\lambda = 1 \Rightarrow x - y + z = 0$ (thrice)

B1

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Any two independent eigenvectors

Attempted

M1

$$\text{(eg)} \quad \alpha \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

A1

$$\lambda = 2 \Rightarrow -y + z = 0$$

$$x - 2y + z = 0 \Rightarrow x = y = z$$

$$x - y = 0$$

M1

$$\gamma \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

A1

5

- (iii) For $\lambda = 1$, eigenvectors represent a plane of invariant points

Plane

M1A1

For $\lambda = 2$, eigenvectors represent an invariant line

B1

3

[20]

Q15.

Attempt at char eqn $\lambda^2 - 7\lambda - 144 = 0$

Any suitable method

Ignore missing “= 0”

M1

Solving quadratic to find evals

Any method

M1

$\lambda = 16$ or -9

CAO

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A1

$$\lambda = 16 \Rightarrow -9x + 12y = 0 \Rightarrow y = \frac{3}{4}x$$

Either λ substituted back

M1

$$\Rightarrow \text{evecs } \alpha \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

CAO (for any non-zero α)

A1

$$\lambda = -9 \Rightarrow 16x + 12y = 0 \Rightarrow y = -\frac{4}{3}x$$

$$\Rightarrow \text{evecs } \beta \begin{bmatrix} 3 \\ -4 \end{bmatrix}$$

CAO (for any non-zero β)

A1

[6]

Q16.

(a) (i) $\det \mathbf{W} = 0$

B1

Transformed shapes have zero volume

*Or equivalent statement
ft volume statements*

B1

2

(ii) Char eqn is $\lambda^3 - 4\lambda^2 + 4\lambda = 0$

One A mark for each coefficient (not the λ^3)

M1A3

Solving the cubic eqn

M1

$\lambda = 0, 2, (2)$

A1

6

Alternative:

$$\begin{aligned}
 \det(\mathbf{W} - \lambda \mathbf{I}) &= \begin{vmatrix} 3-\lambda & -1 & 1 \\ 2 & -\lambda & 2 \\ -1 & 1 & 1-\lambda \end{vmatrix} \\
 &= \begin{vmatrix} 2-\lambda & 0 & 2-\lambda \\ 2 & -\lambda & 2 \\ -1 & 1 & 1-\lambda \end{vmatrix}
 \end{aligned}$$

Use of R/C ops. $R_1' \rightarrow R_1 + R_3$

M1

$$= (2-\lambda) \begin{vmatrix} 1 & 0 & 1 \\ 2 & -\lambda & 2 \\ -1 & 1 & 1-\lambda \end{vmatrix}$$

Factor of $(2-\lambda)$ correctly extracted

A1

$$= (2-\lambda) \begin{vmatrix} 1 & 0 & 0 \\ 2 & -\lambda & 0 \\ -1 & 1 & 2-\lambda \end{vmatrix}$$

$C_1' \rightarrow C_3 - C_1$

$$= (2 - \lambda)^2 \begin{vmatrix} 1 & 0 & 0 \\ 2 & -\lambda & 0 \\ -1 & 1 & 1 \end{vmatrix}$$

A1A1

$$= (2 - \lambda)^2 (-\lambda)$$

Complete factorisation attempt

M1

giving eigenvalues 0 and 2 (twice)

A1

(6)

(b) (i) $x - y + z = 0$

B1

1

(ii)
$$\begin{bmatrix} 3 & -1 & 1 \\ 2 & 0 & 2 \\ -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 3a - b + c \\ 2a + 2c \\ -a + b + c \end{bmatrix}$$

M1A1

$$x' - y' + z' = 3a - b + c - 2a - 2c - a + b + c$$

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M1

$$= 0 \Rightarrow P' \text{ in } H \text{ also}$$

Shown carefully

A1

4

[13]

Q17.

(a) Char. Eqn. is $\lambda^2 - 5\lambda - 6 = 0$

B1

Solving $\Rightarrow \lambda = -1 \text{ or } 6$

M1 A1

Subst^g. either λ back

M1

$$\lambda = -1 \Rightarrow x + y = 0 \Rightarrow \text{evecs. } \alpha \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Any non-zero α

A1

$$\lambda = 6 \Rightarrow 5x - 2y = 0 \Rightarrow \text{evecs. } \beta \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

Any non-zero β

A1

6

(b) (i) $\mathbf{D} = \begin{bmatrix} -1 & 0 \\ 0 & 6 \end{bmatrix} \quad \mathbf{U} = \begin{bmatrix} 1 & 2 \\ -1 & 5 \end{bmatrix}$

ft evals.

B1F

ft evecs. (must correspond to their evals.)

B1F

2

(ii) $\mathbf{U}^{-1} = \frac{1}{7} \begin{bmatrix} 5 & -2 \\ 1 & 1 \end{bmatrix}$

ft their $\mathbf{U}$ (provided non-singular)

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B1F

1

(iii) $\mathbf{X}^5 = \mathbf{U} \mathbf{D}^5 \mathbf{U}^{-1}$

M1

$$= \frac{1}{7} \begin{bmatrix} 1 & 2 \\ -1 & 5 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 6^5 \end{bmatrix} \begin{bmatrix} 5 & -2 \\ 1 & 1 \end{bmatrix}$$

Use of correct $\mathbf{D}^5$ (ft)
N.B. $6^5 = 7776$

B1F

$$= \begin{bmatrix} 2221 & 2222 \\ 5555 & 5554 \end{bmatrix}$$

A1

3

[12]

Q18.

(a) $\mathbf{M} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix}, \mathbf{M} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \\ -2 \end{bmatrix}$
Either/both attempted

Correct

M1

A1A1

$\Rightarrow \lambda_u = 2 \Rightarrow \lambda_v = -2$
If appropriate

B1B1ft

5

(b) $\mathbf{M} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} a - b + c \\ 3a - 3b + c \\ 3a - 5b + 3c \end{bmatrix} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$
Attempt

Correct image

M1

A1

Equating to original + solving

dM1

$\Rightarrow b = c \text{ and } 3a + c = 4b$

A1 A1

Even. is any non-zero multiple of $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

A1

6

(c) (i) $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \mathbf{u} + \mathbf{v} \text{ or } \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

B1

1

(ii) $\mathbf{M}^x \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \mathbf{M}^x (\mathbf{u} + \mathbf{v}) = \mathbf{M}^x \mathbf{u} + \mathbf{M}^x \mathbf{v}$

M1
A1

$$= 2^n \mathbf{u} + (-2)^n \mathbf{v}$$

Replacing M's by evals

M1

$$\lambda = 2 \text{ and } \mu = -2$$

A1

4

(iii)

$$\mathbf{M}^n \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2^n \\ 2^n \\ 2 \times 2^n \end{bmatrix} - \begin{bmatrix} 0 \\ 2^n \\ 2^n \end{bmatrix}$$

Since n is odd (clearly stated)

M1

B1

$$= \begin{bmatrix} 2^n \\ 0 \\ 2^n \end{bmatrix}$$

A1

3

Alternative (a) and (b)

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Char. Eqn. is $k^3 - k^2 - 4k + 4 = 0$

Attempt. One each correct (non-leading) coefft.

M1 A1
A1 A1

Factorisation $(k - 1)(k - 2)(k + 2)$

Attempt

M1

$$k = 1, \pm 2$$

CAO

A1

6

Subst. back any eval.

M1

Solving relevant system(s)

dM1

$$k = 1: \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}; \quad k = 2: \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}; \quad k = -2: \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

Any (non-zero) multiple will do for $k = 1$. For $k = \pm 2$, they must have the evecs. given in the qn. visibly displayed.

A1
A1
A1

[19]

Q19.

(a) (i) Char. Equation is

$$\lambda^2 - 0\lambda + \{-a^2 - (b^2 - a^2)\} = 0$$

Correct to at least here

M1

A1

$$\text{i.e. } \lambda^2 - b^2 = 0 \text{ and } \lambda = \pm b$$

A1

3

(ii) $\lambda = b \Rightarrow (a - b)x + (a + b)y = 0$

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$$\Rightarrow y = \frac{b-a}{b+a}x \Rightarrow \text{eigen vectors } \alpha \begin{bmatrix} b+a \\ b-a \end{bmatrix}$$

AG

A1

2

(iii) $\lambda = -b \Rightarrow (a + b)x + (a + b)y = 0 \text{ (etc.)}$

M1

$$\Rightarrow y = -x \Rightarrow \text{eigen vectors. } \beta \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

A1

2

(b) $\mathbf{D} = \begin{bmatrix} b & 0 \\ 0 & -b \end{bmatrix}, \mathbf{U} = \begin{bmatrix} b+a & 1 \\ b-a & -1 \end{bmatrix}$

B1 B1

$$\text{and } \mathbf{U}^{-1} = \frac{-1}{2b} \begin{bmatrix} -1 & -1 \\ a-b & a+b \end{bmatrix}$$

ft

B1 ✓

$$\mathbf{D}^{11} = \begin{bmatrix} b^{11} & 0 \\ 0 & -b^{11} \end{bmatrix}$$

B1

$$\mathbf{M}^n = \mathbf{U} \mathbf{D}^n \mathbf{U}^{-1} \text{ used}$$

M1

$$\mathbf{M}^{11} = \frac{1}{2} b^{10} \begin{bmatrix} b+a & 1 \\ b-a & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ a-b & a+b \end{bmatrix}$$

or $\frac{1}{2} b^{10} \begin{bmatrix} b+a & -1 \\ b-a & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ b-a & -a-b \end{bmatrix}$

A1

$$= \frac{1}{2} b^{10} \begin{bmatrix} 2a & 2(a+b) \\ 2(b-a) & -2a \end{bmatrix} = b^{10} \mathbf{M}$$

AG

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A1

7

Alternative to (a)(ii)

$$\mathbf{M} \begin{bmatrix} b+a \\ b-a \end{bmatrix} = \begin{bmatrix} ab+a^2+b^2-a^2 \\ b^2-a^2-ab+a^2 \end{bmatrix} = b \begin{bmatrix} b+a \\ b-a \end{bmatrix}$$

M1

A1

Alternative to (b)

$$\text{NB } \mathbf{D}^{11} = b^{10} \mathbf{D}$$

B1

$$\text{Then } \mathbf{M}^{11} = \mathbf{U} \mathbf{D}^{11} \mathbf{U}^{-1}$$

M2 A1

$$= b^{10} \mathbf{U} \mathbf{D} \mathbf{U}^{-1} = b^{10} \mathbf{M}$$

M2 A1

[14]

Q20.

(a) $\begin{bmatrix} 2 & 7 \\ 4 & k \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 9 \\ 4+k \end{bmatrix} = \lambda \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ when $k = 5$,
 $M \times$ given evec.

M1A1

$$\lambda = 9$$

ft

A1

3

(b) Char. Eqn. is $\lambda^2 - 7\lambda - 18 = 0$

M1A1

$(\lambda - 9)(\lambda + 2) = 0$ and 2nd eval. is -2

A1

OR $\det M = \lambda_1 \lambda_2 \Rightarrow -18 = 9\lambda_2$
 $\Rightarrow \lambda_2 = -2$

Or via trace $M = \lambda_1 + \lambda_2$

Subst^g, $\lambda = -2 \Rightarrow 4x + 7y = 0$

M1

$\Rightarrow$ evec. $\begin{bmatrix} 7 \\ -4 \end{bmatrix}$

A1

5

(c) $D = \begin{bmatrix} -2 & 0 \\ 0 & 9 \end{bmatrix}, U = \begin{bmatrix} 7 & 1 \\ -4 & 1 \end{bmatrix}$

ft (alternatives possible)

B1B1

2

(d) $U^{-1} = \frac{1}{11} \begin{bmatrix} 1 & -1 \\ 4 & 7 \end{bmatrix}$

ft non-trivial U's

B1

1

$$(e) \quad \mathbf{M}^{2n} = \begin{bmatrix} 7 & 1 \\ -4 & 1 \end{bmatrix} \begin{bmatrix} (-2)^{2n} & 0 \\ 0 & (9)^{2n} \end{bmatrix} \cdot \frac{1}{11} \begin{bmatrix} 1 & -1 \\ 4 & 7 \end{bmatrix} = \begin{bmatrix} 7 \times 4^n & 81^n \\ -4^{n+1} & 81^n \end{bmatrix} \cdot \frac{1}{11} \begin{bmatrix} 1 & -1 \\ 4 & 7 \end{bmatrix}$$

For D^{2n}

B1

or

$$\begin{bmatrix} 7 & 1 \\ -4 & 1 \end{bmatrix} \cdot \frac{1}{11} \begin{bmatrix} 4^n & -4^n \\ 4 \times 81^n & 7 \times 81^n \end{bmatrix}$$

CAO any correct form

M1
A1

$$\text{Thus } a = \frac{1}{11} \{7 \times 4^n + 4 \times 81^n\}$$

$$\text{i.e. } p = \frac{7}{11}, q = \frac{4}{11}$$

A1

4

In its original form, the question asked for the following conclusion to be made:

Since a is an integer, and $\text{hcf}(4, 11) = 1$,
 $7 \times 4^{n-1} + 81^n$ is a multiple of 11

EXAM PAPERS PRACTICE [15]

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Examiner reports

Q2.

Many students were able to show their understanding of eigenvalues and eigenvectors, scoring full marks in the process. They all approached the question by using the definition of eigenvalues and eigenvectors: $\mathbf{M}\mathbf{v} = \lambda\mathbf{v}$. The unsuccessful students tried in vain to use the characteristic equations getting lost in pages of algebra. The $\mathbf{UDU}^{-1}$ form was well understood by almost all students.

Q3.

The vast majority of students correctly set and expanded the determinant $|\mathbf{M} - \lambda\mathbf{I}| = 0$ although the request to show that $\lambda = 2$ was an eigenvector was sometimes overlooked. Part (b) was well done, although a minority of students mixed up methods and tried to

solve $(\mathbf{M} - 2\mathbf{I}) \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$. In part (c)(i), most students were able to easily verify that $\lambda = 4$ was an eigenvector, although a significant number tried to replicate the method from part (a). Part (c)(ii) proved successful for the majority of students although not all of them realised how the answer could have been obtained from $2 \times 4 = 8$.

Q4.

This question was also generally well done. The only commonly lost mark was that for part (b) which should have been very straightforward; perhaps too straightforward.

Q5.

Apart from the few who did not seem to know what a transpose matrix was, this was a popular and high-scoring question, even for those with relatively low scores on the paper as a whole. The only really disturbing aspect of the work on display was that in part (b)(ii) so few students seemed to have any idea how to go about finding a linear factor of a fairly simple cubic and then factorising to find the accompanying quadratic factor, which is C1 work. Most students who gained the answer $k = -3$ appeared to get it from a calculator, and very few indeed could use it to find the required quadratic factor; fewer still were able to reason convincingly that this quadratic factor had no real roots, which again is C1 work.

In the final part of the question, many students failed to realise that k was now being replaced in the given matrix $\mathbf{A}$ by $k - 7$, so that all they had to do was to add 7 to their previous value of k for two straightforward marks. Almost all students who correctly found the answer for this new k did so by going back, working out a cubic characteristic equation (with some large numbers for coefficients), and using a calculator to solve it for them. This was, somewhat reluctantly, awarded the marks if the answer was correct. Most of those students who did not get the correct answer also used this method, but any incorrect working prevented their calculator helping them out and so they gained no marks.

Q6.

This question produced many more at least partially successful attempts. Nonetheless, sign errors and arithmetical slips abounded and there was, yet again, very little evidence of tracking back to check obviously incorrect working and answers. It was only a very small minority who appreciated that a repeated eigenvalue should lead to an invariant

plane, which needs two representative eigenvectors.

Uncertainty in parts (b) and (c) often led to marks not exceeding 10 out of the 15 available, even amongst the attempts of the better candidates.

Q7.

Candidates generally showed a very good grasp of eigenvalues and eigenvectors in part (a)(i), though some were confused when forming the eigenvectors. The responses to part (a)(ii) were almost equally successful, but some faltered in tackling the inverse of a 2 by 2 matrix, either failing to include the reciprocal of the determinant or incorrectly manipulating the elements of the matrix to form the adjoint matrix.

In part (b)(i), many candidates earned partial credit, either by giving the eigenvalues but failing to show the connection with the diagonalised form, or by doing some good work with the diagonalised form and then failing to state clearly what the eigenvalues were. Part (b)(ii) was answered correctly in many cases but a common error was to write down the meaningless answers ' $\mathbf{v}_1^3$ ' and ' $\mathbf{v}_2^3$ '. Some candidates used the matrix from part (a) in their answers to part (b), but this was accepted as they could still show whether they understood what was needed in part (b).

Q8.

The structure of these questions proved very helpful to candidates, and those who were reasonably careful could generally score at least 16 of this question's 18 marks without any difficulty.

Q9.

This question was handled very well indeed in its technical aspects – writing down eigenvalues and eigenvectors; finding a 2×2 inverse matrix; determining $\mathbf{M}$, and the accompanying matrix multiplications. So a lot of candidates scored lots of marks on this question. However, there were very few candidates indeed who managed to score more than 12 or 13, due to the explanations that were being looked for. At its simplest level, this began in part (a)(i), where candidates were required to write down “the eigenvalues and **corresponding** eigenvectors ...” – in other words, to say which went with which. Examiners did not accept a fortuitously ordered separate pair of each (with a 50-50 chance of a correct pairing). In part (a)(ii), we also wanted to know **why** they had chosen one of the two possible lines as a line of invariant points (as opposed to just an invariant line).

Part (c) had been designed to require thought of the candidates, so the difficulties that arose here were not unexpected. There were two marks for explaining why the given expression was divisible by 13. The first was awarded to anyone who noted that the 1-1 element of $\mathbf{M}^n$ is an integer; the second, subtler, one was for the explanation of why it had to be an integer. Disappointingly few candidates got both marks.

Q10.

The purpose of part (a) was to give candidates a linear factor of the cubic characteristic equation that was to arise in part (b). Most took the hint but then spent far too much time in obtaining this cubic, extracting the linear factor correctly and then solving the resulting quadratic equation. Even amongst those who reached the end successfully, the final ‘simplest surd form’ requirement for the answer caught far too many out.

Q11.

This question required several different ideas. In part (a), it was essential to use the word “area” to describe the significance of $\det \mathbf{M}$ in relation to $\mathbf{T}$, although responses were much more frequently appropriate in this respect than had been the case the last time such a question appeared. A similar type of comment was required at the end of part (b). In part (c), those who parametrised the line as $(x, \frac{1}{2}x + k)$ generally coped very well, although a significant number lost the final mark by not showing the necessary working to support the given answer carefully, as opposed to merely stating what had been stated in the question.

Q12.

The more confident candidates found this question to be a source of easy marks. The real difficulties arose in part (b) due to a widespread inability to use brackets and deal with minus signs. When candidates wrote -3^n when they actually meant $(-3)^n$, and it was necessary to read on to see what they eventually did with the terms involving powers of (-3) . The standard approach appears in the mark scheme, though the alternative algebraic ones also appeared quite often, and these were much more pleasing to the eye. For n even, noting that $\mathbf{D}^n = 3^n \mathbf{I}$, we have $\mathbf{M}^n = 3^n \mathbf{U} \mathbf{I} \mathbf{U}^{-1}$, which gives the required result both obviously and quickly. However, a small number attempted this approach, but wrote something along the lines of $\mathbf{M}^n = \mathbf{U} \mathbf{D}^n \mathbf{U}^{-1} = \mathbf{D}^n \mathbf{U} \mathbf{U}^{-1} = \mathbf{D}^n$ first. This is wrong and

scored 0/3. For n odd, a similar ‘shortcut’ arises via $\mathbf{M}^n = 3^n \mathbf{U} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \mathbf{U}^{-1}$. However, a very small number of candidates are to be highly commended for the following insightful, approach: if n is odd, then $n - 1$ is even, so that $\mathbf{M}^n = \mathbf{M}^{n-1} \mathbf{M} = (3^{n-1} \mathbf{I})(\mathbf{M})$ using the result of part (b)(i) $= 3^{n-1} \mathbf{M}$, as required.

Q13.

This question was actually quite straightforward. However, apart from the final parts of Question 7, it managed to present the biggest obstacle on the paper. Surprisingly, the greatest difficulty arose when candidates were required to turn $(-2)^n$ into 2^n , in the case when n is even, when considering the matrix $\mathbf{D}^n$. Many failed to manage this at all. A disappointingly large number of candidates left it until well into their working and many such candidates then failed to cope with the complicated expressions they were trying to deal with. Those that did so at the outset found themselves with an alternative and simple route through the problem available to them, since it follows that $\mathbf{D}^n = 2^n \mathbf{I}$; it is now straightforward to re-write $\mathbf{T}^n$ and to note that $f(n) = 2^n$.

Q14.

This question was definitely the most demanding one on the paper; partly for its length, and partly due to working with a 3×3 matrix. On top of this, there was a repeated eigenvalue to the matrix, and a lot of candidates clearly did not know quite what to make of it. Nonetheless, even the relatively weaker candidates were still able to gain up to 12 of the marks.

Part (a)(i) was very well handled by those who attempted it. Part (a)(ii) was less well handled. Many of those who fell down here did so because they then went off to find the inverse matrix $\mathbf{M}^{-1}$ itself, rather than follow the logic of the question. Many more who did multiply the matrix equation by $\mathbf{M}^{-1}$ got tangled up in some way – failing to spot that $\mathbf{M} \mathbf{M}^{-1} = \mathbf{I}$, for instance, or being unable to divide by 2 under exam conditions.

In part (b), the factorisation of the determinant $|\mathbf{M} - \lambda \mathbf{I}|$ was usually approached by

row/column operations, which was very pleasing to see. However, it frequently (actually usually) led to the strange sight of candidates taking a partially or completely factorised expression, and then multiplying it out to get a standard cubic form, only to start to factorise it again to find the three eigenvalues.

Those who made a mistake in this working then shot themselves in the foot, losing the following method mark for the attempt to factorise, which they failed to pick up by presuming that their cubic actually was equal to the factorised form $(\lambda - 1)^2(\lambda - 2)$ rather than doing any of the necessary working or going back to check their previous, incorrect working.

The final two parts to the question were demanding, but actually not as difficult as candidates found them to be. Despite the fact that most candidates getting this far quite correctly deduced that $x - y + z = 0$ for the repeated eigenvalue $\lambda = 1$, almost all of them then believed this represented the equation of a line. This error then prevented them from finding a pair of independent representative eigenvectors (or an alternative parametric plane form) and also from describing the result as a plane of invariant points, rather than as a line.

Q15.

This was a routine, straightforward question. A few candidates had difficulty turning a cartesian line equation into an eigenvector, eg $3x = 4y$ became the vector

$$\begin{bmatrix} 3 \\ 4 \end{bmatrix} \text{ rather than } \begin{bmatrix} 4 \\ 3 \end{bmatrix}.$$

Q16.

Incorporating a 3×3 matrix inevitably leads to a higher degree of difficulty in finding the characteristic equation, and this frequently proved to be so for many candidates. Once again, however, it was the introductory remark required to explain the significance of a zero determinant that caused most difficulty. Surprisingly few candidates seemed entirely sure about it, and the most popular suggestion was that $\det \mathbf{W} = 0$ meant that volumes were unchanged. However, the majority of responses dwelt exclusively on the theme of areas, which would have been suitable only had $\mathbf{W}$ been a 2×2 matrix. A very few candidates produced a surprising answer, explaining that $\mathbf{W}$ was a “dimension-destroying (or crushing)” matrix. This gains no credit, unless candidates then go on to explain that 3-d shapes become 2-d ones, or possibly even lines or points, at this stage. A lot of candidates wrote incoherent and meaningless statements: vague statements such as “it is co-planar” don’t carry much weight and certainly don’t get the marks.

Q17.

This question was handled very confidently by most candidates, and apart from a small number of candidates who clearly deduced the matrix $\mathbf{X}^5$ from a calculator, despite the question’s injunction to use the previous results, marks of 10, 11 or 12 were very common indeed.

Q18.

Candidates generally managed the first twelve marks on this question without much difficulty, even if they chose to ignore the question’s structure and go about things in their own way. The final two parts of the paper were deliberately more demanding, requiring candidates to understand that the eigenvalues/eigenvectors of a matrix enable matrix multiplication of certain vectors to be replaced by a scalar multiplication. Amongst those

who seemed to have a good idea of what was going on, the requirement to explain why the final answer was valid for “positive **odd** integers n ” proved beyond them, largely due to a widespread inability to notice that $(-2)^n$ and -2^n are different.

Q19.

The algebraic nature of this question caused many problems, particularly in part (b). For part (a), most candidates had little difficulty in following the standard routine for finding eigenvectors, although many caused themselves problems by failing to simplify the

second eigenvector from $\begin{bmatrix} a+b \\ -b-a \end{bmatrix}$ to $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$. Any correct form was acceptable for the eigenvectors, and if mistakes occurred the candidates could still carry on and earn follow-through marks.

As mentioned above, many candidates did not make any serious attempt at part (b). Of those who did try this part, the overwhelming majority made life particularly hard for themselves by failing to do some fairly simple algebraic processes, such as factorising.

Thus, the product $\begin{bmatrix} b+a & 1 \\ b-a & -1 \end{bmatrix} \begin{bmatrix} b^{11} & 0 \\ 0 & -b^{11} \end{bmatrix} \begin{bmatrix} \frac{1}{2b} & \frac{1}{2b} \\ -\frac{a-b}{2b} & \frac{a+b}{2b} \end{bmatrix}$ could have been made

to look so much easier by taking out the factors of b^{11} and $\frac{1}{2b}$ to get

$$\frac{1}{2}b^{10} \begin{bmatrix} b+a & 1 \\ b-a & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ b-a & -a-b \end{bmatrix}.$$

It was also unfortunate that so many candidates were unable to find the inverse matrix for their choice of U . Some candidates saw that there was no need to find U and U^{-1} at all.

The fact that $D^{11} = b^{10} D$ led them to the reasoning: $M^{11} = U D^{11} U^{-1} = U b^{10} D U^{-1} = b^{10}(U D U^{-1}) = b^{10} M$, though not always expressed quite so concisely.

Q20.

Most candidates managed to score some marks here. Even those picking up 2 or 3 of the marks available here did so retrospectively. The most common error in part (b) was to fail to turn the equation $4x + 7y = 0$, gained from the second eigenvalue of -2 , into a correct eigenvector. Also, the factor of $1/\det U$ was often overlooked. Part (e) was seldom attempted. It was, however, very pleasing to see some completely correct conclusions to the paper.